

PROJECT REPORT

Text Data Mining and Preprocessing

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Abstract

This report details the methodology and results of building a large-scale text dataset from the LessWrong platform. It outlines the specific data collection targets, describes the technical approach for crawling and preprocessing, and presents the final statistics of the compiled datasets.

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1 Target of Data Collection

The objective of this phase is to build a dataset of long-form articles ($\geq 1,500$ words) from the LessWrong platform. After initial collection and cleaning, the data is organized into three distinct datasets with specific volume and time period requirements:

- **cleaned_5**
 - *Target Volume:* 150 articles per author.
 - *Time Period:* Exactly 5 articles published in or after 2025 (≥ 2025), and up to 145 articles published in or before 2024 (≤ 2024).
- **cleaned_10**
 - *Target Volume:* Flexible total, but requires a strict minimum threshold.
 - *Time Period:* A minimum of 20 old articles (≤ 2022) AND a minimum of 20 new articles (≥ 2024). Articles from 2023 are deliberately omitted.
- **cleaned_35**
 - *Target Volume:* Between 30 and 50 articles per author.
 - *Time Period:* No specific year constraints.

2 Methodology

2.1 Data Crawling Process

The data collection was automated using customized Python scripts interfacing directly with LessWrong’s official **graphql api**.

- Running **crawl_lesswrong5.py** populates the dataset enforcing the strict 5/145 temporal split.
- Running **crawl_lesswrong10.py** applies the concurrent ≥ 2024 and ≤ 2022 constraint tracking.
- Running **crawl_lesswrong35.py** collects the general baseline data without strict time constraints.

The crawling infrastructure relies on essential Python libraries:

- **requests:** The core HTTP library used to send POST requests containing JSON-formatted GraphQL queries.
- **BeautifulSoup:** Used to parse the raw `htmlBody` to strip out structural noise (e.g., `<img>`, `<figure>`, `<script>`) and unwrap hyperlinks to extract pure text.
- **re:** Used to enforce the minimum word count accurately. The pattern `\b[\w'-]+\b` ensures words with apostrophes (e.g., "don't") and hyphens are counted correctly as single tokens.

2.2 Further Cleaning

To clean the data more before using it for training, after crawling raw data using 3 files above, the data is cleaned by AI and manually to prevent it from these noises:

- Some articles begin with noises, for example: [This article is reposted from (website)]
- Some posts are deleted by some features such as: audio transcript, post that has excessive quotes or codes or mathjax formulas.

3 Final Dataset Statistics

Following the crawling and cleaning processes, the final datasets were compiled. Detailed tracking metrics for each cleaned directory are presented in the tables below.

Table 1: Detailed data statistics in the `lesswrong5` directory

No.	Author (Slug)	Articles $\geq$ 2025	Articles $\leq$ 2024	Total Articles
1	eliezer_yudkowsky	5	145	150
2	johnswentworth	5	145	150
3	raemon	5	121	126
4	scottalexander	2	145	147
5	zvi	5	145	150
TOTAL		22	701	723

Table 2: Detailed data statistics in the `cleaned_10` directory

No.	Author (Slug)	Articles $\geq$ 2024	Articles $\leq$ 2022	Total Articles
1	abramdemski	20	20	40
2	jessica-liu-taylor	20	20	40
3	johnswentworth	20	20	40
4	kaj_sotala	20	20	40
5	neel-nanda-1	20	20	40
6	raemon	20	20	40
7	ricraz	20	20	40
8	ruby	20	20	40
9	steve2152	20	20	40
10	zvi	20	20	40
TOTAL		200	200	400

Table 3: Detailed data statistics in the `cleaned_35` directory

No.	Author (Slug)	Total Articles
1	abramdemski	45
2	adamshimi	48
3	benito	34
4	benquo	46
5	buck	42
6	david-gross	50
7	eliezer_yudkowsky	46
8	elizabeth-1	36
9	gordon-seidloh-worley	46
10	holdenkarnofsky	38
11	jacob-falkovich	39
12	jan_kulveit	27
13	jasoncrawford	44
14	jessica-liu-taylor	47
15	joe-carlsmith	42
16	johnswentworth	46
17	kaj_sotala	48
18	katjagrace	47
19	lukeprog	38
20	numsempere	27
21	petermccluskey	40
22	philh	29
23	raemon	43
24	ricraz	47
25	ruby	42
26	ryan_greenblatt	44
27	sarahconstantin	49
28	scottalexander	49
29	screwtape	49
30	so8res	40
31	steve2152	48
32	tsvibt	47
33	turntrout	33
34	zack_m_davis	41
35	zvi	47
TOTAL		1484